



ELECTROCHEMISTRY Chapter 12

Teaching Periods

12

Assessment

1

Weightage

8



Students will be able to:

- Give the characteristics of a Redox reaction.
- Determine oxidation and reduction in term of change in oxidation number.
- Determine oxidation number of atom of an element in a molecule or polyatomic ion on the basis of some assigned rules.
- Enlist oxidizing and reducing agents.
- Balance redox reaction that takes place in acid solutions.
- Break a redox reaction into oxidation and reduction half reactions.
- Balance the redox equation by using half-cell reaction method.
- Define cathode, anode, electrode potential and S.H.E.
- Identify the substance oxidized and the substance reduced in batteries.
- Describe the cell potential and how it is determined.
- Describe the reaction that occurs when a lead storage battery is recharged.
- Illustrate how a fuel cell produces electrical energy.
- Explain the types and uses of batteries in daily life.
- Define corrosion and describe simple methods like electroplating and galvanizing for its prevention.

reduced, therefore acts as oxidizing agent. The number of electrons lost and gained must be equal. Various natural processes that we observe in daily routine such as rusting of iron, photosynthesis, fading of cloth's color, generation of electric current in batteries are all concerned with simultaneous reduction and oxidation reactions.

INTRODUCTION

In our daily life many chemical reactions provide electrical energy. Have you ever think how portable radio, children's toys, or an automobile battery works? In fact in all these devices electrical energy generates due to chemical change. In reverse situation, we have a large list of chemical process in which electrical energy is used to bring about chemical changes. For example recharging of battery, extraction and purification of metals etc. Electro chemistry deals with the chemical change happen due to electric current as well as the generation of electric current by chemical change. It is best defined as **"the study of interchange of chemical and electrical energy"**. Electrochemistry is generally investigated by the usage of electrolytic and galvanic cells where electrical energy either produces or utilize by incorporating redox reactions.

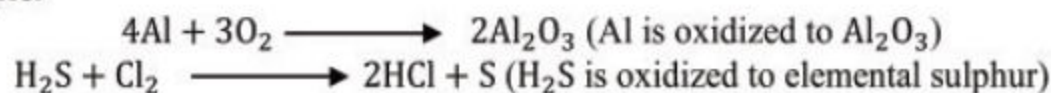
12.1 OXIDATION-REDUCTION CONCEPTS

Reduction and oxidation reactions, collectively known as redox reactions, are associated with the electrons transfer from one species to another. If one species in a reaction loses electrons, it is said to be oxidized and acts as a reducing agent, at the same time the other specie gains electrons and gets

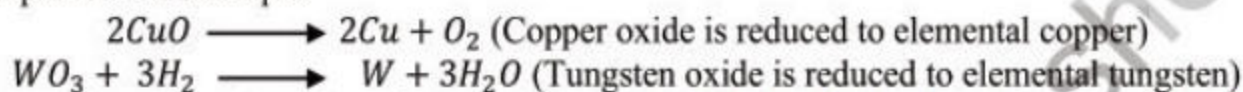


12.1.1 Oxidation and Reduction

Oxidation refers to the addition of oxygen or removal of hydrogen from a substance for example.



Reduction originally describes the addition of hydrogen or removal of oxygen from a compound. For example

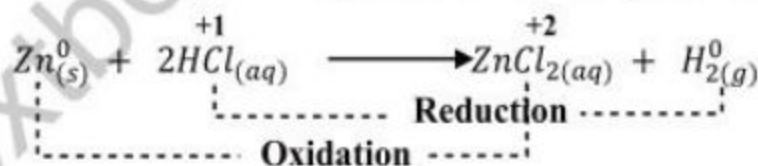


According to modern electronic concept, **“oxidation is the chemical process in which electron are lost by an atom or ion where as reduction involves the gain of electrons”**. Thus oxidation may result in producing or increasing the positive charge on the species or decreasing its negative charge. Contrarily reduction involves in producing or increasing the negative charge on the species or decreasing its positive charge.

Oxidation Process	Reduction Process
$\text{Fe} \longrightarrow \text{Fe}^{+2} + 2\bar{e}$	$\text{Cl}_2 + 2\bar{e} \longrightarrow 2\text{Cl}^-$
$\text{Hg}^+ \longrightarrow \text{Hg}^{+2} + \bar{e}$	$\text{O}^{-1} + \bar{e} \longrightarrow \text{O}^{-2}$
$\text{S}^{-2} \longrightarrow \text{S} + 2\bar{e}$	$\text{Cu}^{+2} + 2\bar{e} \longrightarrow \text{Cu}$

According to oxidation number concept **“oxidation is a process in which the oxidation state of an element increases while reduction is a process in which the oxidation state of an element decreases”**.

Let us consider a redox reaction between zinc powder with dilute hydrochloric acid.



Characteristics of Redox Reactions

There are some general characteristics of redox reactions.

- All redox reactions may involve the transfer of electrons.
- Oxidation and reduction process always occur side by side thus if one substance is oxidized the other is reduced at the same time.
- Not all atoms in a redox reaction need to change their oxidation states but at least one atom should be oxidized and one should be reduced.

12.1.2 Oxidation Number

Oxidation number is an apparent charge on an atom in a compound or ion. Its value may be positive, negative or zero depending upon charge of combined atoms in the molecule or ion.



Rules for Assigning Oxidation Number:

- (i) Oxidation number of all elements in Free State (uncombined state) is taken as zero.
For example: K^0 , Cl_2^0 , H_2^0 , O_2^0 , Cu^0 etc.
- (ii) The oxidation number of a mono atomic ion is the same as the charge appear on it. For example: the oxidation number of Na^+ ion is +1 and Ca^{+2} ion is +2.
- (iii) The sum of oxidation numbers of all elements in a neutral compound is zero. For example: $H_2SO_4^0$ ($H_2^{+2}SO_4^{-2}$)
- (iv) The oxidation number of oxygen in most of its compounds is -2. However in peroxides it is -1 (Na_2O_2 and H_2O_2), in super oxides is -1/2 (KO_2) and in oxy fluoride is +2 (OF_2).
- (v) Oxidation of hydrogen in covalent compounds is assigned as +1. However in metal hydrides its oxidation number is taken as -1 (e.g. Na^+H^- , K^+H^- etc).
- (vi) The oxidation numbers of group IA and group IIA elements are generally assigned +1 and +2 respectively. Elements of group IA are Li, Na, K, Rb, Cs and group IIA are Be, Mg, Ca, Sr, Ba
- (vii) The oxidation number of fluorine in all its compounds is always taken as -1 because it is most electronegative atom of all elements. The oxidation number of other halogen elements in binary compounds is also taken as -1 except when they are bonded to a more electronegative halogen or oxygen.
- (viii) In polyatomic ion, the sum of oxidation numbers of all the atoms present in them is equal to the net ionic charge on the ion.

Example 12.1

Determine the oxidation number of (i) S in $Na_2S_2O_3$ (ii) Mn in MnO_4^-

Solution:

(i) $Na_2S_2O_3$ contains elements Na, S and O. From the rules given above, the oxidation number of Na is +1 and that of O is -2. Further the sum of oxidation number of all elements in a neutral compound is zero. Thus we may determine the oxidation number of sulphur in the following way.

$$2(+1) + 2(S) + 3(-2) = 0$$

$$+2 + 2S - 6 = 0$$

$$2S = +4$$

$$S = +2$$

(ii) Since sum of oxidation number of all the elements in a polyatomic ion is equal to its net ionic charge, we may determine oxidation number of Mn in MnO_4^- ion in the following way.

$$Mn + 4(O) = -1$$

$$(Mn) + 4(-2) = -1$$

$$Mn - 8 = -1$$

$$Mn = +7$$



Self Assessment

Determine the oxidation number in the following species.

- (i) Cl in ClO_3^- (ii) Cr in $\text{Cr}_2(\text{SO}_4)_3$ (iii) P in $\text{Ca}(\text{H}_2\text{PO}_4)_2$

12.1.3 Balancing oxidation reduction equation by ion electron method

Equations of redox reactions are often difficult to balance by simple inspection. There are two methods by virtue of which the redox reactions are balanced, namely; oxidation number method and ion electron method.

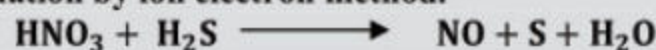
In ion electron method only those reactants and products of redox reaction are balanced that contain the elements undergoing a change in oxidation state. That is to say, only those reactants and products are balanced which are actually oxidized or reduced. Following steps are involved in balancing of chemical equation by ion electron method.

- (1) Write a skeleton equation which includes those reactants and products that contain the elements undergoing a change in oxidation state.
- (2) Transform the molecular skeleton equation into ionic form. If the given equation is already in the ionic form, then this step is omitted.
- (3) Split the ionic equation into two partial equations i.e. the oxidation and reduction equation
- (4) Balance each partial equation in terms of atoms. In neutral or acidic medium, H_2O and H^+ are added for balancing oxygen and hydrogen respectively. The oxygen atoms are balanced first. In case of basic medium, OH^- and H_2O are added to balance oxygen and hydrogen respectively.
- (5) Balance the charge in each partial equation by adding electrons on either left or right side of the equation. It will be found that electrons are added to the left in the partial equation for the reduction equation and to the right in the partial equation for oxidation reaction.
- (6) Multiply each partial equation by a number so that the electrons in both the partial equations become equal in number.
- (7) Add the two partial equations after cancelling the electrons. In the sum equation, cancel out any species common to both sides.



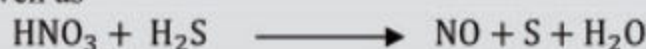
Example 12.2

Balance the following equation by ion electron method.

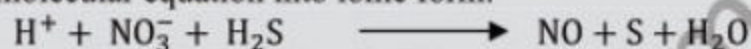


Solution:

The skeleton equation is given as

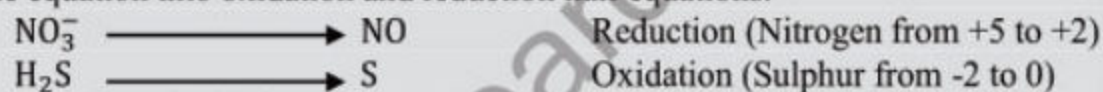


Step-1: Transform the molecular equation into ionic form:



The oxidizing agent is the nitrate ion (NO_3^-) since its nitrogen atom undergoes a decrease in oxidation state. The reducing agent is H_2S since sulphur atom undergoes an increase in oxidation state. H_2S could have been written as sulphide ion (S^{2-}), but H_2S is preferable due to slight degree of ionization.

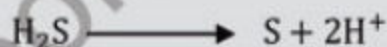
Step-2: Split the equation into oxidation and reduction half equations.



Step-3: Since the medium is acidic, oxygen atoms are balanced by adding $2\text{H}_2\text{O}$ on right hand side and hydrogen atom are balance by 4H^+ on left hand side in the first partial equation.



In the second partial equation two protons are added to the right to balance two hydrogen atoms on the left.



Step-4: Balance the charge in the partial equations by adding electrons.

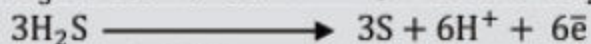
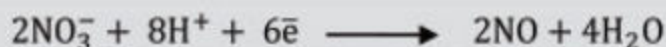
In first partial equation, the oxidation number of nitrogen decreases from +5 to +2, hence 3 electrons are added to the left side.



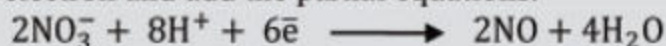
In the second partial equation, the oxidation of sulphur increases from -2 to zero hence 2 electron are added to make it zero on the left



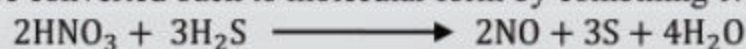
Step-5: In order to equalize the electrons lost and gain, first partial equation is multiplied by 2 and second by 3.



Step-6: Cancel out the electron and add the partial equations.



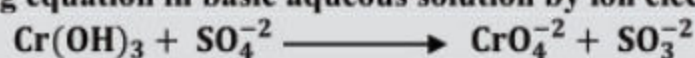
This equation may be converted back to molecular form by combining NO_3^- and H^+





Example 12.3

Balance the following equation in basic aqueous solution by ion electron method.



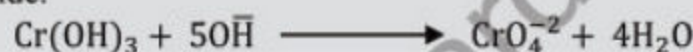
Solution:

To balance a chemical equation, we first identify the elements which undergo change in their oxidation number. In the given equation Cr and S are undergoing oxidation number change.

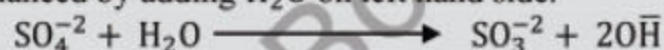
Step-1: Split the given equation into two half reactions mentioning oxidation and reduction on the bases of either the oxidation number of Cr and S decreases or increases.



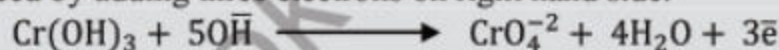
Step-2: According to balancing rule in basic medium, we balance the oxygen atoms of first half equation by adding 5OH^- on left hand side and hydrogen atoms are balanced by adding $4\text{H}_2\text{O}$ on right hand side.



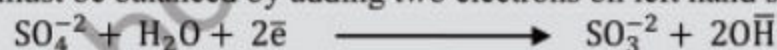
In second half equation, oxygen atoms are balanced by adding 2OH^- on right hand side and hydrogen atoms are balanced by adding H_2O on left hand side.



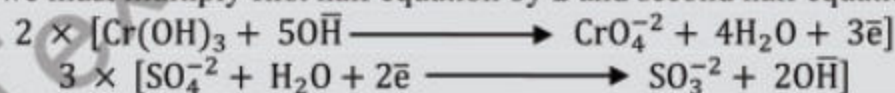
Step-3: Since the oxidation number of Cr increases from +3 to +6, the charges of first half equation are balanced by adding three electrons on right hand side.



On the other hand in second half equation the oxidation number of sulphur decreases from +6 to +4; this change must be balanced by adding two electrons on left hand side.



Step-4: For equalizing the number of electrons lost in first half equation and gain in second half equation, we must multiply first half equation by 2 and second half equation by 3.

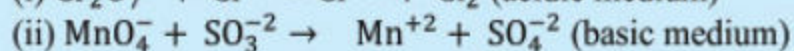
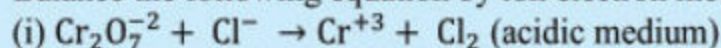


Now add the two half equations and cancel out electrons as well as H_2O in both side to get the net balanced ionic equation.



Self Assessment

Balance the following equation by ion electron method.

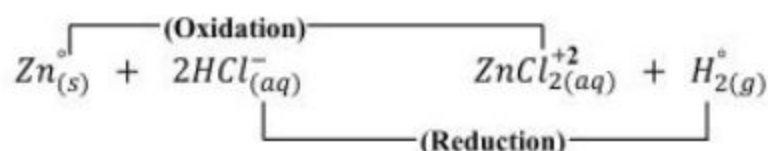




12.1.4 Chemistry of some important oxidizing and reducing agents

Electrons are neither created nor destroyed in a chemical reaction therefore oxidation and reduction always occurs simultaneously.

Consider the oxidation reduction process in the following reaction.



Zinc is one reactant that oxidizes to ZnCl_2 while HCl is other reactant which reduces to H_2 . **“In any redox reaction the species that oxidizes the other substance and itself get reduced is known as oxidizing agent where as the species that reduces other substance but itself get oxidized is identified as reducing agent”.**



Do You Know?

Transmission of nerve impulse in biological system is an electrochemical process and was observed by an Italian scientist Luigi Galvani (1737-1798) during his investigation on frog.

Table 12.1 List of some redox reactions mentioning oxidizing and reducing agent

Redox Reaction	Oxidizing Agent	Reducing Agent
$\text{Cl}_2 + \text{H}_2\text{S} \xrightarrow[\text{Reduction}]{\text{Oxidation}} 2\text{HCl} + \text{S}$	Cl_2	H_2S
$\text{CuO} + \text{H}_2 \xrightarrow[\text{Reduction}]{\text{Oxidation}} \text{Cu} + \text{H}_2\text{O}$	CuO	H_2
$\text{Fe}_2\text{O}_3 + 3\text{CO} \xrightarrow[\text{Reduction}]{\text{Oxidation}} 2\text{Fe} + 3\text{CO}_2$	Fe_2O_3	CO
$\text{Al} + \text{H}_2\text{SO}_4 \xrightarrow[\text{Reduction}]{\text{Oxidation}} \text{Al}_2(\text{SO}_4)_3 + \text{SO}_2 + \text{H}_2\text{O}$	H_2SO_4	Al
$\text{K}_2\text{Cr}_2\text{O}_7 + 6\text{KI} + 7\text{H}_2\text{SO}_4 \xrightarrow[\text{Reduction}]{\text{Oxidation}} \text{Cr}_2(\text{SO}_4)_3 + 4\text{K}_2\text{SO}_4 + 3\text{I}_2 + 7\text{H}_2\text{O}$	$\text{K}_2\text{Cr}_2\text{O}_7$	KI
$2\text{KMnO}_4 + 5\text{H}_2\text{C}_2\text{O}_4 + 3\text{H}_2\text{SO}_4 \xrightarrow[\text{Reduction}]{\text{Oxidation}} 2\text{MnSO}_4 + \text{K}_2\text{SO}_4 + 10\text{CO}_2 + 8\text{H}_2\text{O}$	2KMnO_4	$\text{H}_2\text{C}_2\text{O}_4$



12.2 ELECTRODE, ELECTRODE POTENTIAL AND ELECTROCHEMICAL SERIES

The production or use of electric current through a chemical change is generally carried out in electrochemical cell which is a system consisting of electrodes that dips into an electrolytic solution.

Electrolytic Cell

An electrolytic cell is an apparatus containing two metallic rods or foils known as electrodes dipped in an electrolytic solution and connected to a battery wherein the electric current is used to derive a non-spontaneous redox reaction. The electrode connected to the negative terminal of battery is called cathode, through which electrons enter the solution whereas the electrode connected to the positive terminal of battery is called anode, through which electrons leave the cell. The electrodes are basically metallic plates, rods or foils through which the current enters or leaves the electrolyte. Cathode is the negative electrode (Reductions occurs) whereas anode is the positive electrode (Oxidation occurs). The electrolyte spontaneously dissociates to produce ions within the solution. The positive ions are known as cations while the negative ions are known as anions.

When the electric current is passed through the solution, anions move towards anode where they lose electrons to get oxidized. On the other hand, cations move towards cathode where they gain electrons to become reduced. By this we can say that oxidation occurs at anode while reduction occurs at cathode.

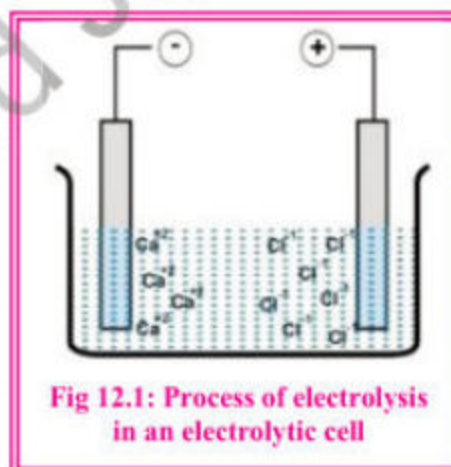
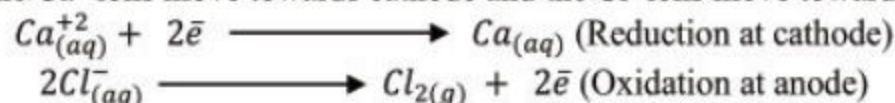


Fig 12.1: Process of electrolysis in an electrolytic cell

“The movement of cations and anions towards their respective electrodes accompanying all chemical changes in an electrolytic solution under the influence of electric current is known as electrolysis”.

During the process of electrolysis, ions of electrolyte migrate towards their respective electrodes where they get oxidized or reduced. For example when aqueous calcium chloride is electrolyzed, the Ca^{2+} ions move towards cathode and the Cl^{-} ions move towards anode.



Thus the reactions taking place in the electrolytic cell are redox reactions and the process of electrolysis remains continue in the electrolytic cell until all ions of the electrolyte change into neutral atoms.



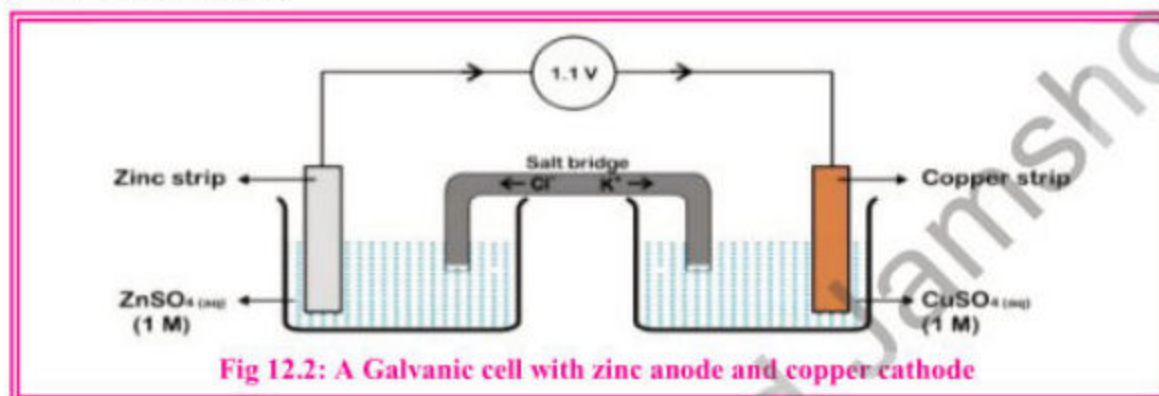
Self Assessment

What is electrolysis? Describe the process of electrolysis for NaCl in an electrolytic cell.



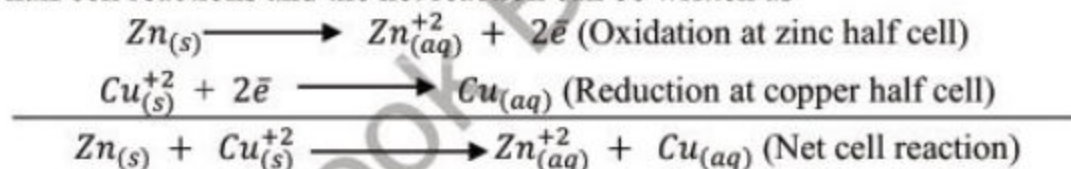
Galvanic Cell

A Galvanic or voltaic cell is type of electrochemical cell in which current is generated by a spontaneous redox reaction. Galvanic cell is typically consist of two half cells which are inter connected electrically. Each half cell is made up of a strip of metal immersed into a solution of its metal ions.

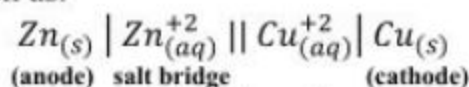


“A salt bridge is a u-shaped glass tube in which an inert electrolyte is filled along with gel. It allows the flow of ions but prevent the mixing of solution”. Thus maintain the electrical neutrality in each half cell.

The two half cell reactions and the net reaction can be written as



Galvanic cells are conveniently express in some short hand notations. For example Zinc-copper cell may be written as.



Here single vertical line specifies the phase boundary between electrode and electrolytic solution where as double vertical lines indicate salt bridge.

Cell Potential

The electrical energy produce by voltaic cell is due to the electron flow from anode to cathode through an external wire. “The driving force with which the electrons pushes out from anode into external circuit is known as electromotive force (emf) or cell potential or cell voltage”.

Just as each half cell reaction is a part of overall reaction, the potential of each half cell make up the overall emf of the cell. Thus, knowing the emf of the cell and electrode potential of any one of the half cell, we could obtained the other. The standard electrode potential of a single half cell is defined as “The difference of potential created between metal electrode and 1 molar solution of its ions at 25°C and 1 atmospheric pressure”. It is abbreviated by E° and measured in volt.

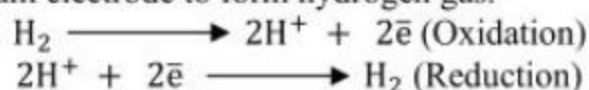


Electrode potential can be represented in term of oxidation potential or reduction potential. The capability of electrode to get oxidized in its electrolytic solution is known as oxidation potential where as the tendency of electrode to gain electrons and allows the positive ions of solution to deposit on it is called as reduction potential. IUPAC convention recommended that electrode potential is taken as reduction potential. Therefore while constructing a Galvanic cell, cathode should be of higher reduction potential and anode should be of lower reduction potential.

It is not possible to measure experimentally the electrode potential of a single electrode. The electrode potential of a single electrode can only be determine if it is coupled with another electrode having a known electrode potential. Chemists established an arbitrary standard known as reference electrode.

The most commonly used reference electrode is **standard hydrogen electrode (SHE)** which assigned as an electrode potential of 0.00 volt and thus the electrode potential of all other electrodes can be measured relative to it.

A standard hydrogen electrode (SHE) is a gas electrode and consists of a platinum rod coated with finely divided platinum which is immersed in 1M solution of H_2SO_4 . Pure hydrogen gas is passed continuously through the solution under a pressure of 1 atm and 25°C . Platinum absorbs hydrogen gas on its surface. When it is connected to another half cell, the hydrogen gas at platinum electrode either pass into solution and form H^+ ions or the H^+ ions of solution reduced at platinum electrode to form hydrogen gas.



If oxidation takes place at platinum electrode, it serves as anode but if reduction occurs, it behaves as cathode.

Standard electrode potential of Galvanic cell (E_{cell}°) is the algebraic difference between the standard reduction potential of two half cells and can be expressed as

$$E_{\text{cell}}^\circ = E_{(\text{cathode})}^\circ - E_{(\text{anode})}^\circ$$

Experimental Determination of Electrode Potential of Zinc and Copper

Standard electrode potential of an electrode can be measured by using a reference electrode such as SHE. Let consider we want to determine the electrode potential of zinc. For this purpose we have to construct a voltaic cell made of Zinc-SHE electrodes. The first half cell consists of a strip of zinc immersed in 1M ZnSO_4 solution while the other half cell consists of platinum foil immersed in 1M H_2SO_4 solution at standard conditions.



Do You Know?

Voltaic cell was first developed by Alessandro volta (1745-1827). He made it by fixing zinc and silver plates alternatively in a container and placed brine soaked card board between them. Since voltmeter was not invented at that time he checked the flow of current by hand shocked.

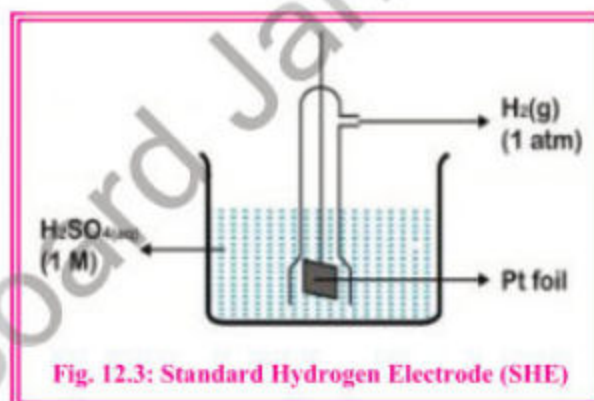


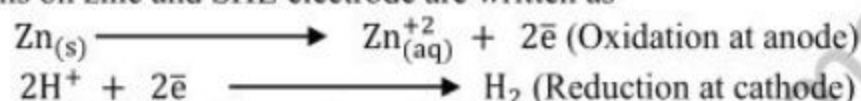
Fig. 12.3: Standard Hydrogen Electrode (SHE)



A salt bridge is made of KCl jelly which completes the circuit between the two half cells and prevent the mixing of solutions. Both electrodes are connected to each other through an external wire by means of a voltmeter.

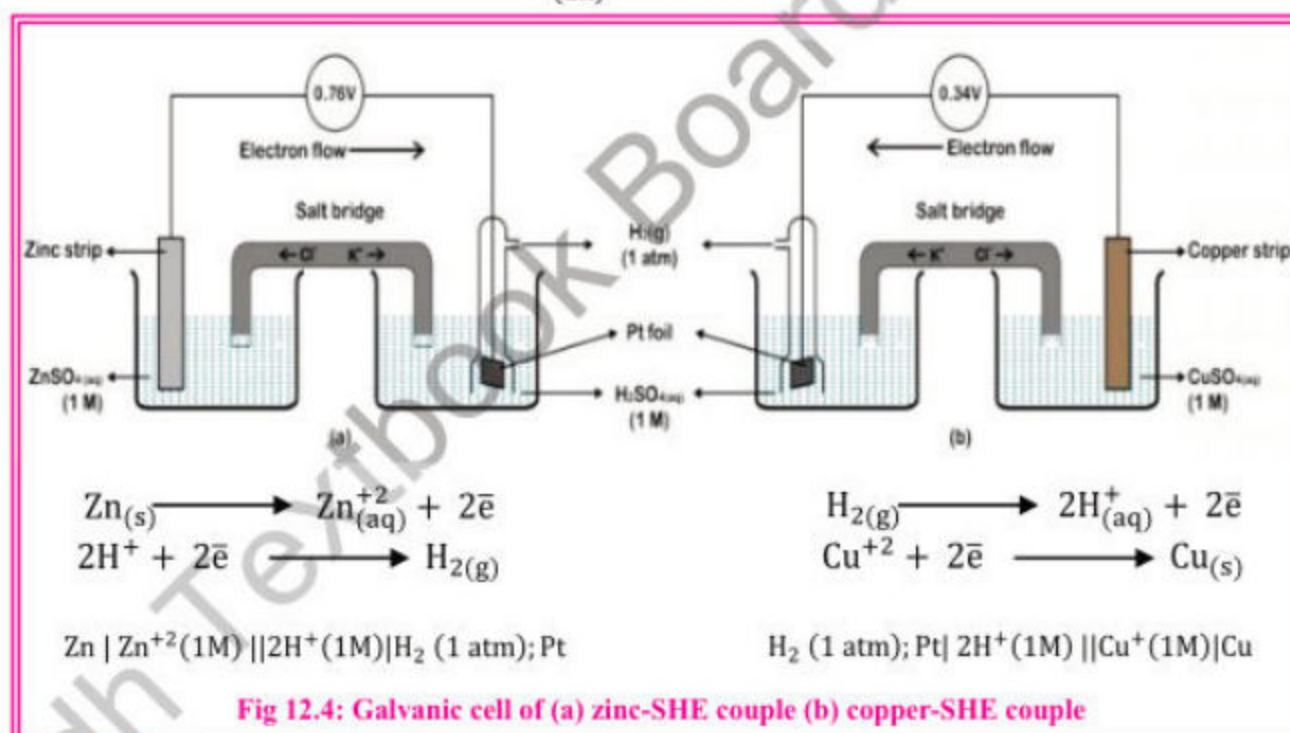
Whenever the construction is completed and Galvanic cell is operational, the voltmeter shows emf of the cell which comes out to be 0.76 volt. Now note the flow of electrons in the external circuit. It should be from zinc electrode to SHE. Thus electrons must have originated at zinc and zinc is anode where oxidation takes place.

Half cell reactions on zinc and SHE electrode are written as



The electrode potential of zinc is now determined as

$$\begin{aligned} E_{(\text{cell})}^{\circ} &= E_{(\text{cathode})}^{\circ} - E_{(\text{anode})}^{\circ} \\ 0.76 &= (0.0) - E_{(\text{Zn})}^{\circ} \\ E_{(\text{Zn})}^{\circ} &= -0.76 \text{ volt} \end{aligned}$$



The electrode potential of copper and other substances can be determined in the same way.

Electrochemical Series

Taking SHE as a reference electrode, chemists determine the electrode potential of many substances and arranged in the order of increasing reduction potential, is known as electro chemical series.

The E° value of each electrode in ECS represents its relative tendency to undergo reduction as compared to the hydrogen ion at standard conditions.

Some important features of electrochemical series are given below:



- (i) Electrodes above hydrogen in ECS have negative electrode potential. They have stronger tendency to lose electron (oxidation) than hydrogen, these substances thus act as reducing agent (anode).
- (ii) Electrodes which are placed below hydrogen in ECS have positive electrode potential. They have greater ability to gain electrons (reduction) as compared to hydrogen and hence serve as oxidizing agent (cathode).

Table 12.2 Standard reduction potential at 25°C for some half reactions			
	Electrodes	Reduction Half – Reaction	E° (V)
Increasing strength of oxidizing agent ↓	Li ⁺ /Li	Li ⁺ + e ⁻ → Li	-3.05
	K ⁺ /K	K ⁺ + e ⁻ → K	-2.93
	Na ⁺ /Na	Na ⁺ + e ⁻ → Na	-2.71
	Mg ²⁺ /Mg	Mg ²⁺ + 2e ⁻ → Mg	-2.37
	Zn ²⁺ /Zn	Zn ²⁺ + 2e ⁻ → Zn	-0.76
	Fe ²⁺ /Fe	Fe ²⁺ + 2e ⁻ → Fe	-0.44
	Cd ²⁺ /Cd	Cd ²⁺ + 2e ⁻ → Cd	-0.40
	Ni ²⁺ /Ni	Ni ²⁺ + 2e ⁻ → Ni	-0.25
	Pb ²⁺ /Pb	Pb ²⁺ + 2e ⁻ → Pb	-0.13
		H ⁺ /H ₂	2H ⁺ + 2e ⁻ → H ₂
↑ Increasing strength of reducing agent	Cu ²⁺ /Cu	Cu ²⁺ + 2e ⁻ → Cu	+0.34
	Ag ⁺ /Ag	Ag ⁺ + e ⁻ → Ag	+0.80
	MnO ₂ /Mn ²⁺	MnO ₂ + 4H ⁺ + 4e ⁻ → Mn ²⁺ + 2H ₂ O	+1.28
	Cr ₂ O ₇ ²⁻ /Cr ³⁺	Cr ₂ O ₇ ²⁻ + 14H ⁺ + 6e ⁻ → 2Cr ³⁺ + 7H ₂ O	+1.33
	Cl ₂ /Cl ⁻	Cl ₂ + 2e ⁻ → 2Cl ⁻	+1.36
	MnO ₄ ⁻ /Mn ²⁺	MnO ₄ ⁻ + 8H ⁺ + 5e ⁻ → Mn ²⁺ + 4H ₂ O	+1.51
	H ₂ O ₂ /H ₂ O	H ₂ O ₂ + 2H ⁺ + 2e ⁻ → 2H ₂ O	+1.78
	F ₂ /2F ⁻	F ₂ + 2e ⁻ → 2F ⁻	+2.87

12.3 BATTERIES

So far we have understood that how a Galvanic cell generate electricity. Indeed the current produce by it is not sufficient for commercial use, practically we combine two or more cells and make a battery.

“A battery consists of two or more Galvanic cells connected in series and is used to convert chemical energy into electrical energy by means of reduction-oxidation (redox) reactions”.

Have you ever seen battery in your automobile car, or the devices like cellular phone, radio, torch, toys etc? You might note that these are light weight, compact in size and work at constant voltage, a commercial battery should fulfill all these criteria.

Batteries are classified into following two types.



12.3.1 Primary Cells/Battery

In this type of batteries redox reaction takes place in only one direction and cannot be reversed. Thus these batteries cannot be recharged or reused. When all amount of electrolyte present in the cell is consumed, they become dead. The most inexpensive and commonly used primary cell is dry cell which we often use in remote control, transistors, flash lights and many other domestic works.

Dry Cell

The common dry cell was first invented by a French chemist Leclanche. The outer body of this cell is made up of zinc which serves as anode where as a graphite rod located at the centre works as cathode. The whole zinc container is lined with porous paper which works as a separator. The region between cathode and anode is filled with a paste of MnO_2 , Carbon, NH_4Cl and ZnCl_2 .

When the cell starts working Zn anode oxidizes to Zn^{+2} ions whereas at graphite cathode Mn^{+4} ions reduces to Mn^{+3} ions.



The electrons generated at anode flow out to the external circuit. This cell produces 1.5 volt electric current and become dead after complete utilizing zinc electrode. 0



Do You Know?

Technologically, electrochemistry plays an imperative role in many ways in our lives. For example

(i) Batteries that power a variety of appliances vehicles and devices become more compact light weight and consumable day by day due to the research progress in electrochemistry.

(ii) Other domains of modern science in which electrochemistry is significantly involves are electro refining, electroplating, corrosion, protection, electrophoresis etc.

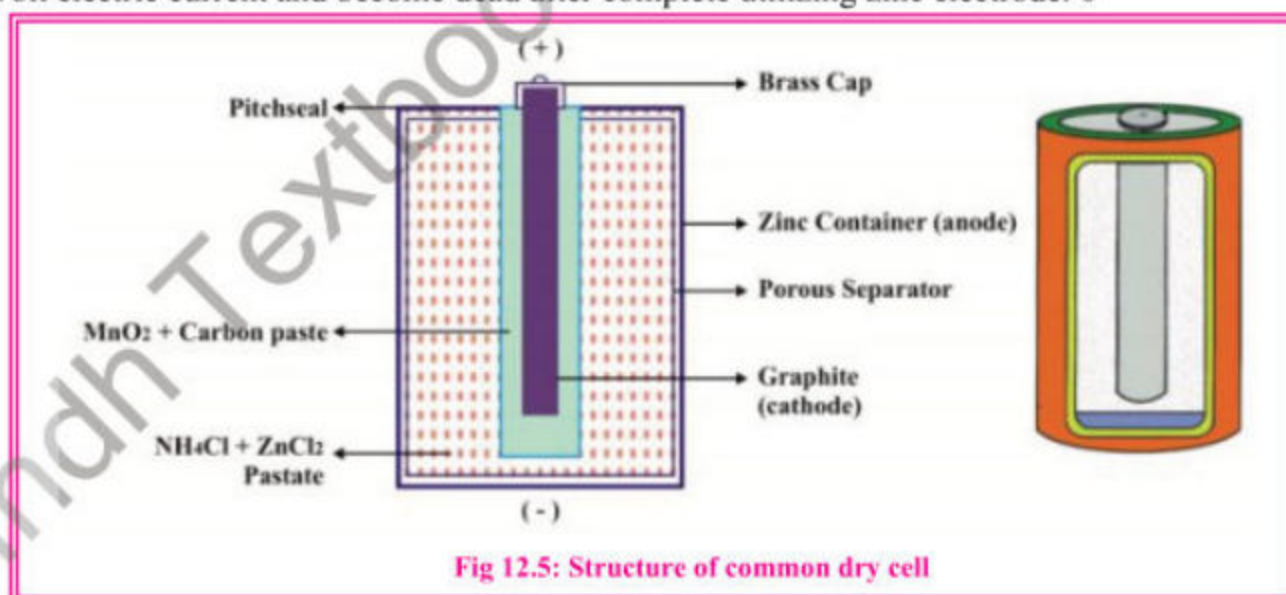


Fig 12.5: Structure of common dry cell

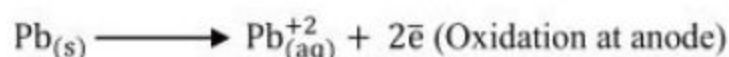
12.3.2 Secondary Battery

These batteries can be recharged by reversing the electricity flow through the cell by utilizing a source of external power supply after every use. Lead storage battery which is commonly known as car battery is an example of secondary battery.



A 12 volt car battery is made up of six Galvanic cells connected in a series each of which has an emf of 2 volt. All anode plates are made up lead (Pb) while all cathode plates consists of lead coated with PbO_2 . Both groups of plates are alternatively suspended into a dilute solution of H_2SO_4 of specific gravity 1.25.

When the battery starts working, lead atoms at anode oxidize to form lead ions.



The Pb^{+2} ions combine with SO_4^{-2} ions to form solid PbSO_4 which is deposited on anode.



The electrons generate at anode pushes out into external circuit and then re-enter into cell from cathode (PbO_2) where PbO_2 reduces to Pb^{+2} ions.



The Pb^{+2} ions again combine with SO_4^{-2} ions of electrolyte and produce solid PbSO_4 which is deposited at cathode plates.

The net chemical change at anode and cathode during discharge may be written as.

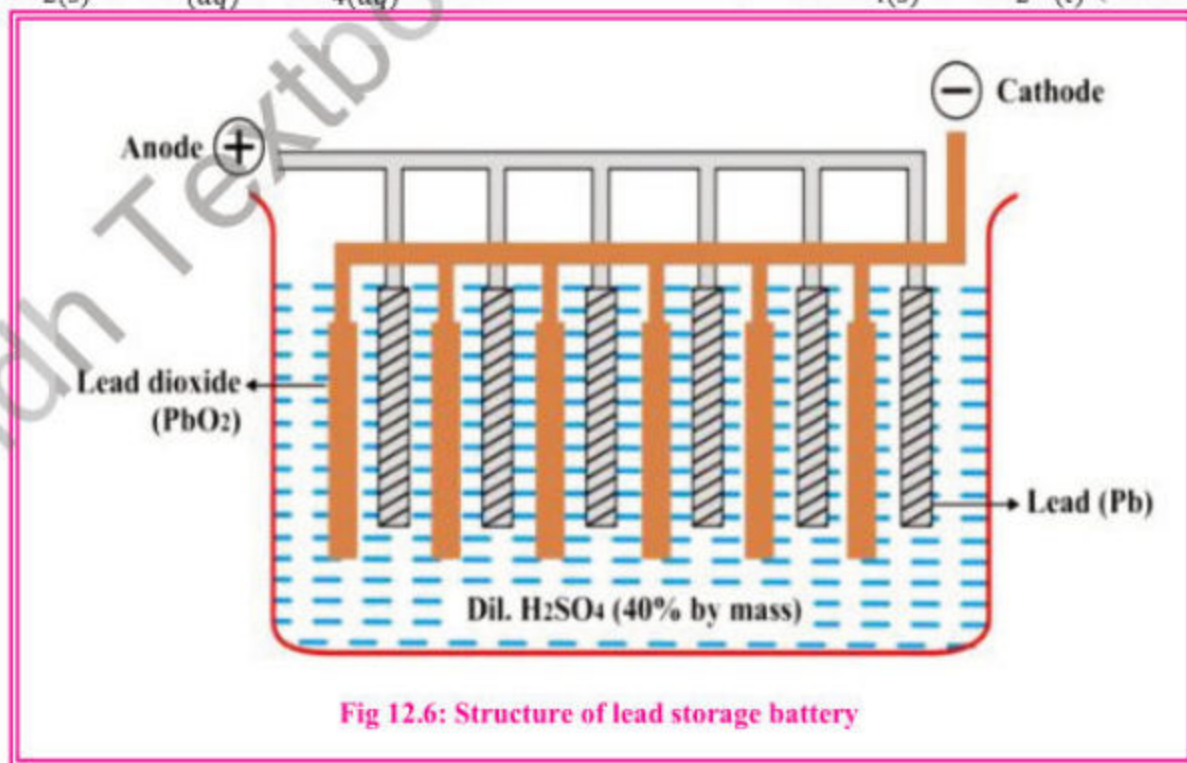
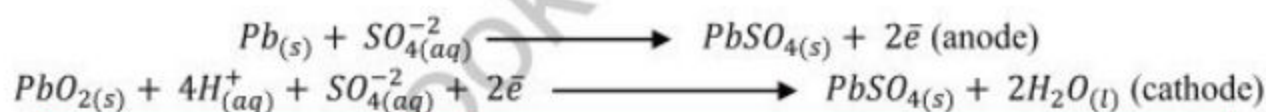


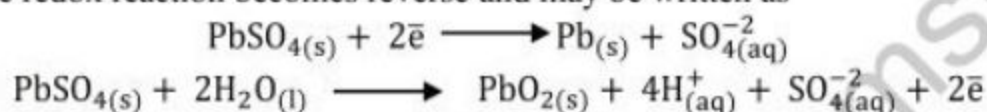
Fig 12.6: Structure of lead storage battery



As long as lead battery works, its electrolyte (H_2SO_4) uses up gradually which results in covering the electrodes with PbSO_4 . Thus it becomes unable to release electricity and said to be discharge. This can be checked by lowering in the specific gravity of acid as well as drop of voltage.

How is car battery recharged?

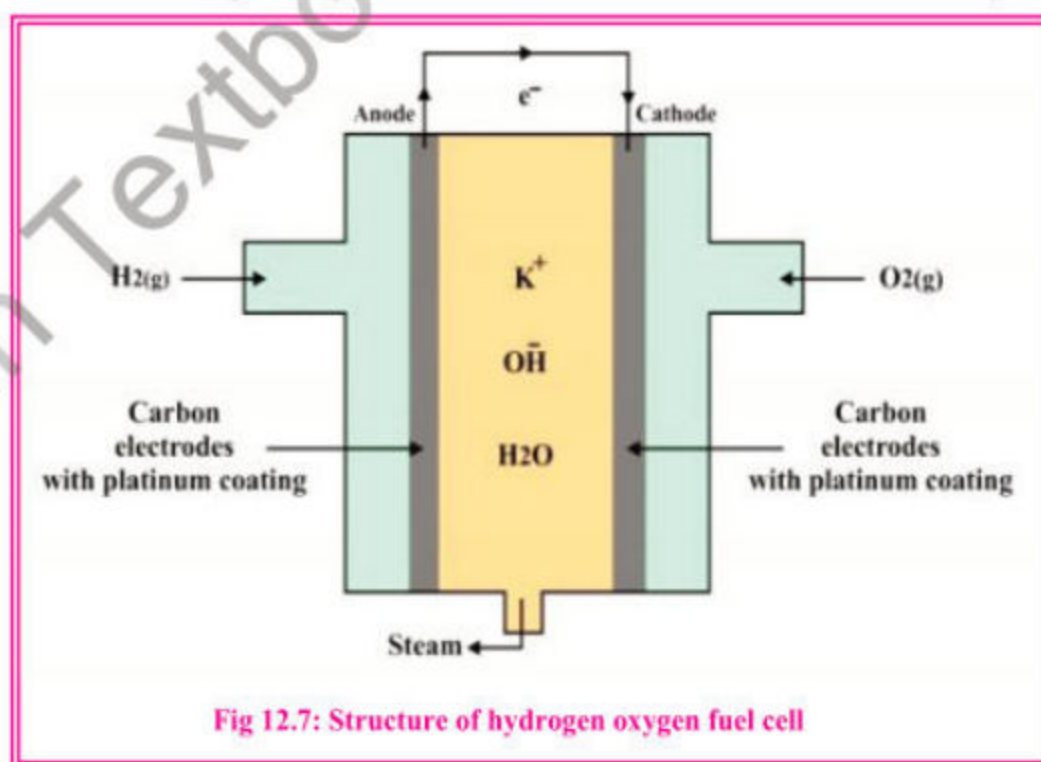
When the battery is needed to be recharged, its anode (Pb) is connected to negative terminal and cathode (PbO_2) is connected to positive terminal of external power supply. In doing so, the redox reaction becomes reverse and may be written as



Thus H^{+} ions conc. increases gradually in the electrolyte until the specific gravity again reaches the original value and the voltage of the battery is restored.

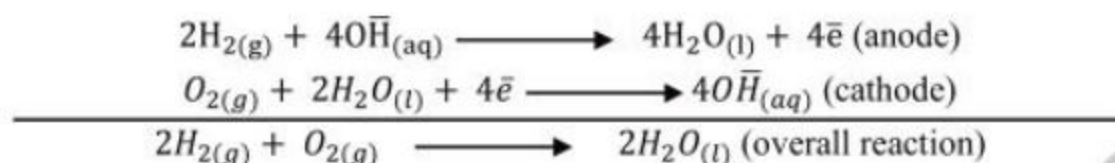
12.3.3 Fuel Cell

A hydrogen oxygen fuel cell is unique type of Galvanic cell which is based on the reaction between hydrogen and oxygen to produce water and the heat release in the reaction is use to produce electricity. Unlike other batteries, it cannot store reactants hence reactants are fed continuously which utilize in the reaction and the product thus formed remove out continuously. The most familiar Fuel cell is hydrogen oxygen fuel cell. It consists of two porous carbon electrodes having a thin platinum paste. Platinum serves as catalyst. The compartment between the two electrodes is filled with concentration solution of KOH which works as electrolyte. Hydrogen gas fed into the cell from anode side and oxygen from cathode side. These gases after bubbling through their concerned electrode enter into electrolytic solution.





The reaction at cathode and anode are given below:



12.4 CORROSION AND ITS PREVENTION

We often noted that metals on exposure to air lose their luster and strength. A brown, green or grayish layer develops on their surface which makes them dull. It is due to chemical interaction between metal and the environment such as oxygen of air and moisture. **“A spontaneous process in which surface atoms of metals get oxidized (harmful oxides) due to the action of surrounding medium is known as corrosion”**. Generally, corrosion occurs on the surface of metal but when the upper surface gets corroded, the atoms in the inner layer get exposed to air and become corroded. This process remains continue up to a certain depth which makes the metal brittle, and weak. However some metals are self protective to corrosion since only the atoms of outer layer oxidize.

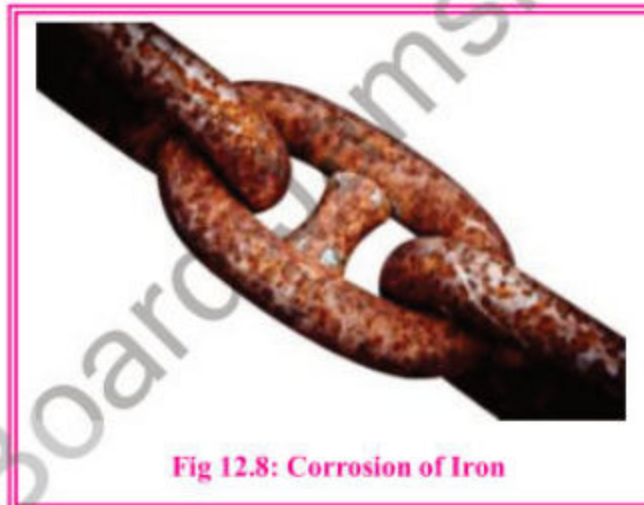


Fig 12.8: Corrosion of Iron

The most familiar type of corrosion which we see in daily life is the **“rusting of iron”**. Rust is a brown powdery material produce on the surface of iron appliances due to an oxidation reaction of iron atoms with moist air.



Various theories describes the phenomena of corrosion but the most adequate is the electrochemical theory which tells that metal atoms serves as cathode and anode and water containing O_2 and CO_2 works as electrolyte which involves in the flow of electrons. In the light of this theory **“corrosion is an electrochemical process”**.

Prevention against corrosion

We aware that iron, copper and various other metals are essentially used in many commercial as well as domestic purpose. To keep them for long period, they should be protected against corrosion by using certain temporary or permanent techniques. Iron Gate of our house is painted so that paint resists the contact of moist air with the metal atoms. Coating of grease or oil in machineries and tools are commonly seen for corrosion control. Often a thin layer of another metal such as zinc, tin or chromium is coated on the metal. For more permanent safety, alloying of metals is carried out.



Galvanizing

“The process of coating a thin layer of zinc on a metal is known as galvanizing”. Since zinc is placed above iron in ECS, it preferably oxidizes and consumes rather than iron and thus protects the iron against corrosion. Galvanizing is carried out either by spraying molten zinc on the surface of iron made material or by dipping the iron sheets into molten zinc.

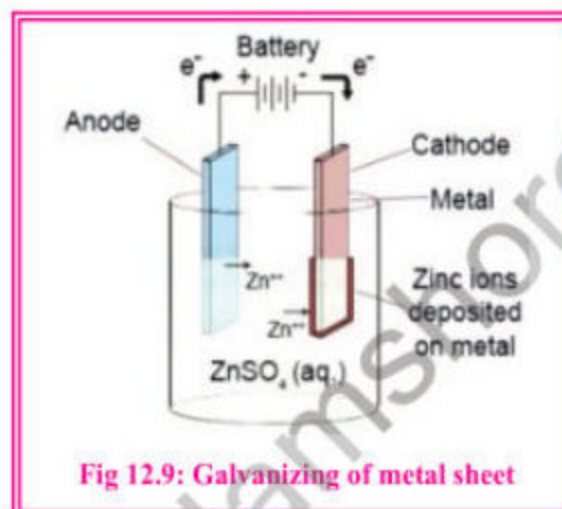


Fig 12.9: Galvanizing of metal sheet

Electroplating

We have studied in sec 12.2 that during the process of electrolysis metal cat ions of electrolytic solution deposited on cathode. Thus a thin layer of tin, chromium etc can be coated on given metal, this process is known as electroplating. Its most common use is electroplating of tin. **“Electroplating is the coating of one metal on the surface of another by involving electrolysis”.** Vegetable oil, food and juice containers are generally tin plated. Tin plating protects iron as long as it remains coated, if scratched then iron exposed to air and starts to get corrosion.

In the electrolytic process, the iron sheets after cleaning, washing and drying are suspended into a bath containing a mixture of stannous chloride and hydrochloric acid. Pure tin is made the anode and the iron sheet works as cathode. On passing the current through the bath, tin starts depositing on the iron sheet and thus iron sheets are tinned electrolytically.



Society, Technology and Science

Solar cell as a source of energy

Solar cell or photovoltaic cell is a device which converts solar energy into useable electricity. The material of most of the solar cell is silicon which serves as semiconductor. When exposed to sun light, it absorbs radiations and transfer energy of photons into electrons of silicons. The electrons after absorbing this energy jump out into the external circuit.

Solar technology provides us an opportunity to make saving in electricity bills. Although the cost of solar panels installation is high but it produces free electricity. Further, remote areas which are beyond electric power transmission, this method of renewable energy becomes popular day by day.



Activity

The combination of copper and aluminum strips in the citric acid of the lemon serves as the source of electric energy in this activity.

Just insert aluminum foil and copper coin in the lemon such that they remain as far as possible from each other. Now connect these metals to a bulb with the help of wires.

The lemon's citric acid serves as an electrolyte. Since aluminum is located in the upper side of ECS, so it oxidizes and releases electrons to the external circuit. Copper on the other hand, undergoes reduction by gaining the electrons and thus bulb is lit by an electric current.

SUMMARY with Key Terms

- ◆ **Oxidation** is a chemical process in which electrons are lost by an atom or ion. During this process oxidation number of an element increases.
- ◆ **Reduction** is a chemical process in which electrons are gained by an atom or ion. During this process oxidation number of an element decreases.
- ◆ **Oxidizing agent** is the specie which oxidizes the other substance and itself get reduced in a redox reaction. Stronger oxidizing agents found in the lower region of ECS. Thus strongest oxidizing agent is fluorine.
- ◆ **Reducing agent** is the specie which reduces the other substance but itself get oxidized in a redox reaction. Stronger reducing agent is located on upper region of ECS. Thus strongest reducing agent is Li.
- ◆ **Redox reactions** are associated with electron transfer from one species to another. Each redox reaction can be broken into two half reactions, one comprises with the loss of electron and other with the gain of electrons.
- ◆ **Electrochemical cell** is a system in which electric current is either producing or use through chemical change.
- ◆ **Electrolysis** is the process in which a non spontaneous reaction takes place at the expense of electric energy. During this process ions of electrolyte migrate towards their respective electrodes where they get oxidized or reduced.



- ◆ **Galvanic cell** is a device in which electricity is generated by spontaneous redox reaction. It consists of two half cells and each is made up of a strip of metal immersed into a solution of its metal ions. Cathode in this cell is positive and anode is negative.
- ◆ **Salt bridge** is a U-shaped glass tube in which an inert electrolyte is filled along with gel. It allows the flow of ions but prevents the mixing of solution.
- ◆ **Cell Potential (E_{cell}°)** of a Galvanic cell is the driving force with which the electrons push out from anode into external circuit. It is measured by connecting two half cells through a voltmeter.
- ◆ **Standard electrode potential (E°)** is the difference of potential created between a metal electrode and one molar solution of its ions at 25°C and 1 atmospheric pressure.
- ◆ **Standard hydrogen electrode (SHE)** is a reference electrode which is assigned as electrode potential of 0.0 volt. It consists of a platinum rod immersed in 1M acid solution where pure hydrogen gas at 1 atmospheric pressure and 25°C is passed.
- ◆ **Electrochemical series (ECS)** is an arrangement of various electrode potentials in order of their increasing values. All the electrodes above hydrogen in ECS have negative values and they are reducing agents, whereas below hydrogen electrode potential values are positive indicating that they are oxidizing agents.
- ◆ **Corrosion** is an electrochemical process in which atoms on the metal surface get oxidized due to environmental oxygen and moisture.
- ◆ **Galvanizing** is the coating of a thin layer of zinc on the metal to protect it from corrosion.
- ◆ **Electro tinning** is the coating of tin on metal sheet by using the process of electrolysis. It helps in preventing a metal from corrosion.
- ◆ **Battery** is the source of direct current and is constructed by connecting two or more galvanic cells.
- ◆ **Primary batteries/Cells** are not rechargeable; thus when their chemicals have been consumed, they stop doing chemical reactions.
- ◆ **Secondary batteries/Cells** are galvanic cells in which chemical reactions are easily being reversible. These cells can be recharged by flowing current in reverse direction.
- ◆ **Dry Cell** is a primary cell in which metallic zinc is oxidized to Zn^{+2} ions and liberates electrons which flow along the container to the external circuit.



- ◆ **Lead storage battery** is a portable secondary battery use in automobile in which lead serves as anode and lead oxide (PbO_2) works as cathode.
- ◆ **Fuel Cell** is a special type of Galvanic cell which generates electricity by chemical reactions but a continuous supply of reactants and a continuous removal of product are required.

EXERCISE

Multiple Choice Questions

1. Choose the correct answer

- (i) The outer body of dry cell serves as anode, it is made up of:
(a) Copper (b) Zinc
(c) Lead (d) Iron
- (ii) The conduction of electricity through an electrolytic solution is due to the flow of:
(a) Electrons (b) Ions
(c) Atoms (d) Molecules
- (iii) During electrolysis, the reaction that takes place at anode is:
(a) Oxidation (b) Reduction
(c) Hydrolysis (d) Simultaneous oxidation and reduction
- (iv) This statement is not correct for lead storage battery:
(a) It can be recharged (b) It is a primary battery
(c) Anode is made up of lead (d) Cathode is made up of lead oxide
- (v) In Zn-SHE voltaic cell, the half reaction occurs at anode is:
(a) $\text{Zn}^{+2} + 2\bar{e} \longrightarrow \text{Zn}$ (b) $\text{Zn} \longrightarrow \text{Zn}^{+2} + 2\bar{e}$
(c) $2\text{H}^+ + 2\bar{e} \longrightarrow \text{H}_2$ (d) $\text{H}_2 \longrightarrow 2\text{H}^+ + 2\bar{e}$
- (vi) The strongest oxidizing agent in the electro chemical series is:
(a) Li (b) H_2
(c) Cu (d) F_2
- (vii) Galvanized rode of iron is coated with:
(a) Nickel (b) Zinc
(c) Chromium (d) Carbon
- (viii) Fuel cell is a typical Galvanic cell which is based on the reaction between:
(a) Nitrogen and oxygen (b) Hydrogen and oxygen
(c) Methane and oxygen (d) Hydrogen and zinc
- (ix) Oxidation number of Cr in $\text{Na}_2\text{Cr}_2\text{O}_7$ is:
(a) +3 (b) +6
(c) +8 (d) +12
- (x) Which of the following half cell reaction show oxidation:
(a) $\text{Fe}^{+3} \rightarrow \text{Fe}^{+2}$ (b) $\text{Cl}_2 \rightarrow 2\text{Cl}^-$
(c) $\text{SO}_4^{-2} \rightarrow \text{SO}_4^{-3}$ (d) $\text{Zn} \rightarrow \text{Zn}^{+2}$



Short Questions

1. Define the following:
 - Primary and secondary cell
 - Oxidizing agent and reducing agent
 - Redox reactions
 - Oxidation and reduction
 - Oxidation number
2. What is meant by oxidation number? Determine the oxidation number in following.
 - (i) Cr in H_2CrO_4
 - (ii) S in $\text{K}_2\text{S}_4\text{O}_6$
 - (iii) Fe in Fe_3O_4
 - (iv) Cl in HClO_4
3. Define the term standard electrode potential and hydrogen electrode.
4. In the process of electroplating the item to be plated is made cathode, give reason?
5. What is corrosion? What causes it to form? What can be done to prevent its formation?
6. How can you define an electrochemical series? Give its properties.

Descriptive Questions

1. (a) What is meant by electrode potential and standard Hydrogen electrode (SHE).
(b) Describe how the electrode potential of copper is determined.
2. Sketch a copper-zinc Galvanic cell, write the cell reaction and show the direction of electron flow.
3. What is the difference between a primary cell and secondary cell sketch a diagram of dry cell and explain its working.
4. Balance the following equations by ion electron method.
 - (i) $\text{Cr}_2\text{O}_7^{2-} + \text{I}_2 \longrightarrow \text{Cr}^{+3} + \text{IO}_3^-$ (acidic medium)
 - (ii) $\text{MnO}_4^- + \text{SO}_3^{2-} \longrightarrow \text{Mn}^{+2} + \text{SO}_4^{2-}$ (basic medium)
 - (iii) $\text{MnO}_4^- + \text{Cl}^- \longrightarrow \text{Mn}^{+2} + \text{Cl}_2$ (acidic medium)
 - (iv) $\text{Cl}_2 + \text{OH}^- \longrightarrow \text{Cl}^- + \text{ClO}_3^- + \text{H}_2\text{O}$ (basic medium)
 - (v) $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \longrightarrow \text{Mn}^{+2} + \text{CO}_2 + \text{H}_2\text{O}$ (acidic medium)
 - (vi) $\text{Fe}^{+2} + \text{Cr}_2\text{O}_7^{2-} + \text{H}^+ \longrightarrow \text{Fe}^{+3} + \text{Cr}^{+3} + \text{H}_2\text{O}$ (acidic medium)



MCQS ANSWER KEYS

Chap #	MCQs No.									
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
1	c	d	a	a	a	c	a	b	b	b
2	b	c	d	d	c	b	b	c	a	a
3	c	b	d	a	c	c	c	a	d	c
4	d	b	c	c	b	d	d	d	a	c
5	d	a	c	b	d	b	a	b	a	a
6	d	c	c	c	b	d	a	c	a	c
7	a	c	d	b	d	a	d	c	a	a
8	c	c	d	d	c	a	c	d	b	c
9	c	a	c	c	a	c	b	d	b	b
10	d	c	c	d	b	b	d	d	c	b
11	d	b	c	c	c	a	a	c	c	a
12	b	b	a	b	b	d	b	b	b	d



APPENDIX - 1

Multiples and Submultiples of SI Units

Prefixes	Symbols	Multiplying Factor	
tera	T	10^{12}	1 000 000 000 000
giga	G	10^9	1 000 000 000
mega	M	10^6	1 000 000
kilo	K	10^3	1 000
hecto	H	10^2	100
deca	Da	10	10
deci	D	10^{-1}	0.1
centi	C	10^{-2}	0.01
milli	M	10^{-3}	0.001
micro	U	10^{-6}	0.000 001
nano	n	10^{-9}	0.000 000 001
pico	p	10^{-12}	0.000 000 000 001



APPENDIX - 2

Atomic Numbers & Atomic Weights

Element	Symbol	Atomic Number	Atomic Weight
Hydrogen	H	1	1.008
Helium	He	2	4.003
Lithium	Li	3	6.938
Beryllium	Be	4	9.0121
Boron	B	5	10.821
Carbon	C	6	12.0116
Nitrogen	N	7	14.007
Oxygen	O	8	15.999
Fluorine	F	9	18.998
Neon	Ne	10	20.179
Sodium	Na	11	22.989
Magnesium	Mg	12	24.307
Aluminum (aluminium)	Al	13	26.9815385
Silicon	Si	14	28.086
Phosphorus	P	15	30.973
Sulfur (sulphur)	S	16	32.059
Chlorine	Cl	17	35.446
Argon	Ar	18	39.948
Potassium	K	19	39.098
Calcium	Ca	20	40.078



Element	Symbol	Atomic Number	Atomic Weight
Scandium	Sc	21	44.955
Titanium	Ti	22	47.867
Vanadium	V	23	50.941
Chromium	Cr	24	51.996
Manganese	Mn	25	54.938
Iron	Fe	26	55.845
Cobalt	Co	27	58.933
Nickel	Ni	28	58.693
Copper	Cu	29	63.546
Zinc	Zn	30	65.380
Gallium	Ga	31	69.723
Germanium	Ge	32	72.630
Arsenic	As	33	74.921
Selenium	Se	34	78.971
Bromine	Br	35	79.907
Krypton	Kr	36	83.798
Rubidium	Rb	37	85.467
Strontium	Sr	38	87.62
Yttrium	Y	39	88.905
Zirconium	Zr	40	91.224



Element	Symbol	Atomic Number	Atomic Weight
Niobium	Nb	41	92.906
Molybdenum	Mo	42	95.950
Technetium	Tc	43	97
Ruthenium	Ru	44	101.07
Rhodium	Rh	45	102.905
Palladium	Pd	46	106.42
Silver	Ag	47	107.868
Cadmium	Cd	48	112.414
Indium	In	49	114.818
Tin	Sn	50	118.710
Antimony	Sb	51	121.760
Tellurium	Te	52	127.600
Iodine	I	53	126.904
Xenon	Xe	54	131.293
Cesium (caesium)	Cs	55	132.905
Barium	Ba	56	137.327
Lanthanum	La	57	138.905
Cerium	Ce	58	140.116
Praseodymium	Pr	59	140.907
Neodymium	Nd	60	144.242



Element	Symbol	Atomic Number	Atomic Weight
Promethium	Pm	61	145
Samarium	Sm	62	150.36
Europium	Eu	63	151.964
Gadolinium	Gd	64	157.25
Terbium	Tb	65	158.925
Dysprosium	Dy	66	162.500
Holmium	Ho	67	164.930
Erbium	Er	68	167.259
Thulium	Tm	69	168.934
Ytterbium	Yb	70	173.045
Lutetium	Lu	71	174.966
Hafnium	Hf	72	178.49
Tantalum	Ta	73	180.947
Tungsten (wolfram)	W	74	183.84
Rhenium	Re	75	186.207
Osmium	Os	76	190.23
Iridium	Ir	77	192.217
Platinum	Pt	78	195.084
Gold	Au	79	196.966
Mercury	Hg	80	200.592



Element	Symbol	Atomic Number	Atomic Weight
Thallium	Tl	81	204.385
Lead	Pb	82	207.2
Bismuth	Bi	83	208.980
Polonium	Po	84	209
Astatine	At	85	210
Radon	Rn	86	222
Francium	Fr	87	223
Radium	Ra	88	226
Actinium	Ac	89	227
Thorium	Th	90	232.037
Protactinium	Pa	91	231.035
Uranium	U	92	238.028
Neptunium	Np	93	237
Plutonium	Pu	94	244
Americium	Am	95	243
Curium	Cm	96	247
Berkelium	Bk	97	247
Californium	Cf	98	251
Einsteinium	Es	99	252
Fermium	Fm	100	257



Element	Symbol	Atomic Number	Atomic Weight
Mendelevium	Md	101	258
Nobelium	No	102	259
Lawrencium	Lr	103	262
Rutherfordium	Rf	104	263
Dubnium	Db	105	268
Seaborgium	Sg	106	271
Bohrium	Bh	107	270
Hassium	Hs	108	270
Meitnerium	Mt	109	278
Darmstadtium	Ds	110	281
Roentgenium	Rg	111	281
Copernicium	Cn	112	285
Ununtrium	Uut	113	286
Flerovium	Fl	114	289
Ununpentium	Uup	115	289
Livermorium	Lv	116	293
Ununseptium	Uus	117	294
Ununoctium	Uuo	118	294



APPENDIX - 3

Constant Values

Physical Constant	Symbols	Values
Speed of light	C	$3 \times 10^8 \text{ m/s}$
Planck constant	H	$6.625 \times 10^{-34} \text{ J.S}$
Boltzmann constant	K	$1.38 \times 10^{-23} \text{ J/K}$
Universal gas constant	R	8.314 J/(mol.K) $0.082 \text{ atm dm}^3/\text{mol.K}$
Avogadro's number	N_A	6.023×10^{23}
Charge of electron	E	$1.602 \times 10^{-19} \text{ Coulomb}$
Permittivity of vacuum	ϵ_0	$8.84 \times 10^{-12} \text{ C}^2/\text{J.m}$
Mass of electron	m_e	$9.1 \times 10^{-31} \text{ kg}$
Mass of proton	m_p	$1.6726 \times 10^{-27} \text{ kg}$
Mass of neutron	m_n	$1.6749 \times 10^{-27} \text{ kg}$
Rydberg constant	R_h	$1.09678 \times 10^7 \text{ m}^{-1}$
Bohr radius	a_0	$0.529 \times 10^{-10} \text{ m}$
Molar Volume	$V(\text{STP})$	22.4 dm^3
Standard Temperature	$T(\text{STP})$	$273 \text{ K} = 0^\circ\text{C}$
Standard Pressure	$P(\text{STP})$	$101325 \text{ Pa} = 1 \text{ atm}$
Standard Hydrogen Electrode	SHE	0 volt
Density of water at 4°C	$\rho_{\text{H}_2\text{O}}$	1 g/cm^3
Specific heat of water	$C_{\text{H}_2\text{O}}$	4200 J/Kg.K
Ionization constant of water	$K_W(125^\circ\text{C})$	1×10^{-14}

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS

1 H Hydrogen 1.008																	2 He Helium 4.003																								
3 Li Lithium 6.941	4 Be Beryllium 9.012									5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180																										
11 Na Sodium 22.990	12 Mg Magnesium 24.305							13 Al Aluminum 26.982	14 Si Silicon 28.086	15 P Phosphorus 30.974	16 S Sulfur 32.066	17 Cl Chlorine 35.453	18 Ar Argon 39.948																												
19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.88	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.631	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798																								
37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium 98.907	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.906	46 Pd Palladium 106.42	47 Ag Silver 107.868	48 Cd Cadmium 112.414	49 In Indium 114.818	50 Sn Tin 118.711	51 Sb Antimony 121.760	52 Te Tellurium 127.6	53 I Iodine 126.904	54 Xe Xenon 131.294																								
55 Cs Cesium 132.905	56 Ba Barium 137.328											57 La Lanthanum 138.905	58 Ce Cerium 140.12	59 Pr Praseodymium 140.908	60 Nd Neodymium 144.24	61 Pm Promethium [145]	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97															
87 Fr Francium [223]	88 Ra Radium 226.025											89 Ac Actinium [227]	90 Th Thorium 232.038	91 Pa Protactinium 231.036	92 U Uranium 238.029	93 Np Neptunium 237.048	94 Pu Plutonium 244.064	95 Am Americium 243.061	96 Cm Curium 247.070	97 Bk Berkelium 247.070	98 Cf Californium 251.083	99 Es Einsteinium [252]	100 Fm Fermium [257]	101 Md Mendelevium [258]	102 No Nobelium [259]	103 Lr Lawrencium [260]	104 Rf Rutherfordium [261]	105 Db Dubnium [262]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [265]	109 Mt Meitnerium [268]	110 Rg Roentgenium [269]	111 Ds Darmstadtium [271]	112 Cn Copernicium [285]	113 Nh Nihonium [286]	114 Fl Flerovium [289]	115 Mc Moscovium [288]	116 Lv Livermorium [293]	117 Ts Tennessine [294]	118 Og Oganesson [294]

87	La	Lanthanum	138.905
88	Ce	Cerium	140.116
89	Pr	Praseodymium	140.908
90	Nd	Neodymium	144.242
91	Pm	Promethium	144.913
92	Sm	Samarium	150.36
93	Eu	Europium	151.964
94	Gd	Gadolinium	157.25
95	Tb	Terbium	158.925
96	Dy	Dysprosium	162.50
97	Ho	Holmium	164.930
98	Er	Erbium	167.259
99	Tm	Thulium	168.934
100	Yb	Ytterbium	173.054
101	Lu	Lutetium	174.967
102	No	Nobelium	259.101
103	Lr	Lawrencium	[260]
104	Rf	Rutherfordium	[261]
105	Db	Dubnium	[262]
106	Sg	Seaborgium	[266]
107	Bh	Bohrium	[264]
108	Hs	Hassium	[265]
109	Mt	Meitnerium	[268]
110	Rg	Roentgenium	[269]
111	Ds	Darmstadtium	[271]
112	Cn	Copernicium	[285]
113	Nh	Nihonium	[286]
114	Fl	Flerovium	[289]
115	Mc	Moscovium	[288]
116	Lv	Livermorium	[293]
117	Ts	Tennessine	[294]
118	Og	Oganesson	[294]

Alkali Metal	Alkaline Earth	Transition Metal	Basic Metal	Metalloid	Nonmetal	Halogens	Noble Gas	Lanthanide	Actinide
--------------	----------------	------------------	-------------	-----------	----------	----------	-----------	------------	----------

